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TREND-BASED MOCK PAPER

GATE Mathematics

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mathematics
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Mathematics

Q1. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q2. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q3. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q4. [1-NAT] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q5. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms

- (B) Cosine terms
- (C) Both
- (D) Neither

Q6. [1-NAT] [1 mark]

The vector (1,2,3) projected onto (2,1,0) is:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q8. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q9. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q10. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q11. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q12. [1-NAT] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite

- (C) Diagonally dominant
- (D) Non-singular

Q14. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q15. [1-NAT] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-NAT] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q17. [1-NAT] [1 mark]

Laplace transform of $e^{(-3t)} \cdot u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q20. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q21. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q22. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68

- (B) 0.32
- (C) 0.95
- (D) 0.16

Q23. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q25. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q26. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q27. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q28. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q29. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q30. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q31. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q32. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2, 1 \\ 1, 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
 - (B) $[2, 1]^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q33. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C * x$
 - (B) $y = C/x$
 - (C) $y = C * x^2$
 - (D) $x^2 + y^2 = C$
-

Q34. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C * x$
 - (B) $y = C/x$
 - (C) $y = C * x^2$
 - (D) $x^2 + y^2 = C$
-

Q36. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q37. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q38. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q39. [1-NAT] [1 mark]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q40. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

Q41. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q42. [2-NAT] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q43. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \times 2$
 - (D) $P \times r/100$
-

Q44. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-NAT] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q46. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q47. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q48. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q49. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q50. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q51. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q52. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q53. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MSQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$

(D) $P \cdot r / 100$

Q55. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q56. [1-NAT] [1 mark]

Identity element property:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

Rank of 3×3 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q58. [1-NAT] [1 mark]

Inverse of $AB = ?$

(Numerical Answer Type — Enter the numerical value)

Q59. [1-NAT] [1 mark]

Ring must have:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-NAT] [1 mark]

Order of $y'' + 3y' + 2y = 0$:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-NAT] [1 mark]

Limit of $\sin(x)/x$ as $x \rightarrow 0$:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q63. [1-MCQ] [1 mark]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q64. [1-MCQ] [1 mark]

Normal distribution PDF:

- (A) $(1/\sigma\sqrt{2\pi})\exp(-(x-\mu)^2/(2\sigma^2))$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q65. [1-MCQ] [1 mark]

$f(x)=|x|$ at $x=0$:

- (A) Discontinuous
 - (B) Continuous
 - (C) Differentiable
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	A	1
2	1MCQ	D	1
3	1NAT	A	1
4	1NAT	C	1
5	1NAT	C	1
6	1NAT	B	1
7	1MCQ	B	1
8	1MCQ	A	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1NAT	A	1
12	1MCQ	A	1
13	1MCQ	A	1

14	1MCQ	C	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1NAT	A	1
18	1MCQ	A	1
19	1NAT	B	1
20	1MCQ	A	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	A	1
24	1MCQ	A	1
25	1NAT	B	1
26	1MCQ	C	1
27	1MCQ	B	1
28	1NAT	C	1

29	1NAT	B	1
30	1NAT	A	1
31	2NAT	A	2
32	2MCQ	C	2
33	2MCQ	A	2
34	2MCQ	C	2
35	2MCQ	A	2
36	2NAT	A	2
37	2MCQ	B	2
38	2MCQ	A	2
39	2NAT	A	2
40	2MCQ	A	2
41	2MCQ	C	2
42	2NAT	A	2
43	2MCQ	A	2

44	2NAT	A	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2MCQ	A	2
50	2MCQ	A	2
51	2MCQ	A	2
52	2NAT	C	2
53	2MCQ	A	2
54	2MCQ	A	2
55	2MCQ	A	2
56	2MCQ	A	2
57	2NAT	A	2
58	2MCQ	B	2

59	2MSQ	A	2
60	2NAT	A	2
61	2MCQ	B	2
62	2MCQ	A	2
63	2MCQ	B	2
64	2MCQ	A	2
65	2NAT	A	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Group Theory | Difficulty: difficult

Q2.

Answer: D

Solution:

Correct Answer: D

Subject: Algebra | Topic: Linear Algebra | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Matrix | Difficulty: easy

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Algebra | Topic: Ring Theory | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Calculus | Topic: ODE | Difficulty: difficult

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Calculus | Topic: Real Analysis | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Analysis | Topic: Metric Spaces | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Topology | Topic: Open Sets | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Probability | Topic: Distributions | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Real Analysis | Topic: Continuity | Difficulty: difficult

Q11.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q12.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q13.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q14.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q15.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q16.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q18.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q19.

Answer: B

Solution:

$\text{proj} = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q20.

Answer: A

Solution:

$|\langle u,v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q22.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q25.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q40.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q41.

Answer: C

Solution:

$y = A*\cos(2x) + B*\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q43.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q45.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q48.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q49.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q50.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q51.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q52.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{(n-2)} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q55.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q56.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q57.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q58.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q60.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q65.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 38 questions, 59 marks
Miscellaneous: 17 questions, 31 marks
Algebra: 4 questions, 4 marks
Calculus: 2 questions, 2 marks
Analysis: 1 questions, 1 marks
Topology: 1 questions, 1 marks
Probability: 1 questions, 1 marks
Real Analysis: 1 questions, 1 marks

Difficulty Distribution:

Easy: 19 | Moderate: 36 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.